

ERSIS — Enterprise Retail & Strategic Inventory System

Logical Design: List of Tables and Foreign Keys

Key:

Bold + Underline = Primary Key (PK) Underline = Foreign Keys (FK) Plain = Regular Attribute

1. STORE MANAGEMENT

- **STORES** (store_id, store_name, owner_id, address, contact_email, contact_phone, is_active, created_at);
- **CATEGORIES** (category_id, store_id, category_name, parent_category_id, description);

2. AUTH & SECURITY

- **ROLES** (role_id, role_name, description, created_at);
- **USERS** (user_id, username, email, password_hash, first_name, last_name, phone, is_active, created_at, updated_at);
- **USER_ROLES** (user_role_id, user_id, role_id, store_id, is_active, assigned_at, revoked_at);
- **OTP_TOKENS** (token_id, user_id, otp_code_hash, purpose, expires_at, is_used, created_at);
- **REFRESH_TOKENS** (token_id, user_id, token_hash, device_info, ip_address, expires_at, revoked_at, created_at);
- **USER_DEVICE_TOKENS** (token_id, user_id, device_token, platform, is_active, last_used_at, created_at);

3. PRODUCT CATALOG

- **PRODUCTS** (product_id, store_id, category_id, product_name, barcode, description, unit_price, tax_rate, unit_of_measure, is_active, created_at, updated_at);
- **INVENTORY** (inventory_id, product_id, store_id, quantity_in_stock, reorder_level, last_restocked_at, updated_at);
- **SUPPLIERS** (supplier_id, supplier_name, contact_person, email, phone, address, created_at);
- **PRODUCT_SUPPLIERS** (product_supplier_id, product_id, supplier_id, supply_price, lead_time_days, is_preferred);
- **PRODUCT_PRICE_HISTORY** (price_history_id, product_id, old_price, new_price, changed_by, reason, changed_at);
- **PRODUCT_IMAGES** (image_id, product_id, image_url, alt_text, is_primary, display_order, uploaded_by, created_at);

4. SALES & TRANSACTIONS

- **GUEST_CUSTOMERS** (guest_id, name, phone, email, created_at);
- **DISCOUNTS** (discount_id, store_id, discount_name, discount_type, discount_value, applies_to, product_id, category_id, min_purchase_amount, valid_from, valid_until, is_active, created_at);
- **TRANSACTIONS** (transaction_id, invoice_number, store_id, cashier_id, customer_id, guest_customer_id, transaction_date, subtotal, tax_amount, discount_amount, total_amount, status, created_at, updated_at);
- **TRANSACTION_ITEMS** (item_id, transaction_id, product_id, quantity, unit_price_at_sale, discount_id, discount, line_total);
- **TRANSACTION_DISCOUNTS** (id, transaction_id, discount_id, applied_amount, applied_at);
- **REFUND_ITEMS** (refund_id, original_transaction_id, refund_transaction_id, product_id, quantity_returned, refund_amount, reason, notes, processed_by, created_at);
- **PAYMENTS** (payment_id, transaction_id, payment_method, amount, payment_status, payment_reference, paid_at);

5. INVENTORY & IOT

- **INVENTORY_LOGS** (log_id, inventory_id, product_id, store_id, movement_type, quantity_change, quantity_before, quantity_after, reference_type, reference_id, notes, performed_by, created_at);
- **IOT_DEVICES** (device_id, store_id, device_identifier, device_type, mqtt_username, is_active, registered_at, last_seen_at);
- **SCAN_LOGS** (scan_id, device_id, barcode_value, scan_timestamp, transaction_id, status);
- **SALES_FORECASTS** (forecast_id, store_id, product_id, forecast_date, predicted_quantity, rmse_score, mae_score, model_version, generated_at);

6. PURCHASE ORDERS

- **PURCHASE_ORDERS** (order_id, store_id, supplier_id, ordered_by, status, order_date, expected_date, received_date, notes, created_at);
- **PURCHASE_ORDER_ITEMS** (item_id, order_id, product_id, quantity_ordered, unit_cost, quantity_received);

7. AI, ANALYTICS & AUDIT

- **CHATBOT_SESSIONS** (session_id, user_id, store_id, access_level, started_at, ended_at);
- **CHATBOT_MESSAGES** (message_id, session_id, sender_type, message_text, retrieved_context, sent_at);
- **AUDIT_LOGS** (log_id, user_id, store_id, action, entity_type, entity_id, old_value, new_value, ip_address, created_at);

- **RAG_DOCUMENT_CHUNKS** (chunk id, store id, source_type, access_level, source_id, chunk_text, faiss_index_id, embedding_model, is_active, created_at, updated_at);
- **STORE_FAQS** (faq id, store id, question, answer, is_active, created_at, updated_at);
- **STORE_POLICIES** (policy id, store id, policy_name, content, access_level, is_active, created_at, updated_at);
- **NOTIFICATIONS** (notification id, user id, store id, notification_type, channel, subject, body, status, reference_type, reference_id, sent_at, read_at, created_at);